PRASANTH RAYUDU

ASIC VERIFICATION ENGINEER | SYSTEMVERILOG | FPGA & SOC VALIDATION

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PROFESSIONAL SUMMARY

Computer Engineering M.S. candidate with hands-on experience developing SystemVerilog/Verilog testbenches, reusable bus-functional models, FPGA emulation flows, regression automation, and hardware-software co-validation for SoC and Edge-AI systems. Experienced writing Python, embedded C, Tcl, Make, and Bash tooling; troubleshooting RTL-to-hardware mismatches; analyzing waveforms and VCD activity; and validating datapaths, interfaces, timing, power, and performance. Strong foundation in verification methodology, functional debug, DFT/scan/BIST, ModelSim, Synopsys VCS, Cadence Xcelium, Vivado, Xilinx ILA, AXI, PCIe, I2C, LVDS, ARM embedded Linux, and firmware-facing hardware interactions. Seeking to contribute to NVIDIA GPU and SoC verification as a new-college-graduate engineer.

CORE TECHNICAL SKILLS

RTL & Digital Design: Verilog, SystemVerilog, VHDL, RTL design, datapaths, ALU/MAC units, control FSMs, digital logic, microarchitecture, clock gating

Validation & Debug: RTL simulation, functional validation, testbench development, regression testing, pre-silicon validation, FPGA emulation, waveform analysis, failure triage, CDC analysis/debug

Verification & DFT: SystemVerilog testbenches, reusable verification components, bus-functional models, transactors, functional debug, DFT scan, BIST, ATPG, fault simulation

FPGA & Systems: Xilinx Vivado, ModelSim, Synopsys VCS, Xilinx ILA, UltraScale/UltraScale+, MPSoC, ARM-based systems, embedded Linux, hardware-software co-validation

Power, Performance & PPA: Dynamic power estimation, VCD activity parsing, switching-factor analysis, power/performance/area tradeoffs, resource utilization, performance-per-watt, voltage/frequency scaling, workload benchmarking

Edge AI & Memory: AI/ML accelerators, HDC, compute-in-memory, RRAM/PCM non-idealities, quantization, pruning, ADC/macro tradeoffs, hardware-software co-design

Programming & Validation Tooling: Python, C, C++, embedded C, MATLAB, Tcl, Bash/Shell, Make, CMake, Git, CI/CD, log parsing, data processing, Linux systems

ASIC/SoC & Hardware Debug: ASIC/SoC verification concepts, firmware-facing validation, troubleshooting, on-chip logic analyzer debug, waveform analysis, CDC, timing closure, FPGA place-and-route, Cadence Xcelium

Interfaces: AXI, PCIe, I2C, LVDS, memory-mapped I/O, interrupts, hardware-software co-simulation

EDUCATION

Master of Science in Computer Engineering | San Diego State University **| Expected Dec 2026**

- Relevant coursework: Computer Architecture, VLSI Design, Digital Systems, Embedded Systems, Memory Systems, Low-Power Microarchitecture, Multi-Core Execution, Cache and Memory Hierarchies

Bachelor of Technology in Electronics and Communication Engineering | VR Siddhartha Engineering College

PROFESSIONAL EXPERIENCE

Graduate Research Assistant – Hardware-Software Co-Design for Edge AI | San Diego State University **| Aug 2025 – Present**

- Develop HyperOPT, a Python-based hardware-software co-design framework for exploring compute-in-memory and hyperdimensional-computing accelerator configurations under accuracy, energy, latency, and area constraints.
- Model quantization, analog noise, memory-cell non-idealities, ADC overhead, and multi-macro mapping to evaluate performance-per-watt, accuracy, and implementation tradeoffs for edge-AI workloads.
- Implement pruning and binary-search strategies that reduce design-space exploration cost while producing reproducible benchmark reports and CSV outputs for architecture and design reviews.

FPGA Emulation and Validation Flow Automation | Engineering Project **| Jan 2024 – Aug 2025**

- Built a pre-silicon FPGA emulation and validation flow with CI-triggered compilation, regression execution, test generation, and simulation-to-hardware handoff on Xilinx UltraScale-class platforms.
- Developed Python, Tcl, Make, and Bash automation for test launching, environment configuration, workload execution, metric collection, log parsing, and repeatable validation artifacts.
- Added failure clustering and analysis that enabled root-cause triage for more than 95% of simulation-to-emulation mismatches and reduced compile/debug iteration time by 30%.

Teaching Associate – General Studies | San Diego State University **| Aug 2024 – May 2025**

- Led weekly seminar sections, communicated complex technical concepts to diverse audiences, and collaborated in teaching communities to improve course execution and student support.

Student Intern – FPGA and Machine Learning Systems | In22 Labs **| Jun 2021 – Jul 2022**

- Designed and validated Verilog RTL blocks for a biomedical machine-learning pipeline and co-simulated embedded C reference models to verify functional correctness.
- Automated ModelSim waveform and log validation in Python, reducing manual debug effort by 50% and improving regression reproducibility for FPGA-oriented development.

SELECTED ENGINEERING PROJECTS

RTL and FPGA Validation of a Custom Execution Datapath

- Designed a custom execution datapath with control logic and ALU-style math blocks in Verilog/SystemVerilog, applying microarchitecture and digital-logic fundamentals from specification through RTL implementation.
- Resolved clock-domain crossing issues, performed static timing analysis, and achieved timing closure on a Xilinx MPSoC FPGA prototype using Vivado.
- Collaborated with architecture and verification teams during design reviews to evaluate implementation tradeoffs and meet performance and power targets across compute workloads.

System-Level FPGA Emulation and SoC Validation Framework

- Integrated a multi-IP CPU-microcontroller subsystem on a Xilinx MPSoC FPGA for pre-silicon validation and ARM embedded-Linux bring-up, evaluating system behavior and integration tradeoffs.
- Created reusable SystemVerilog testbenches, bus-functional models, and transactors for AXI memory-mapped traffic, interrupts, PCIe, I2C, and LVDS interfaces; reduced build cycles by 40%.
- Used Synopsys VCS, ModelSim, Vivado, and Xilinx ILA to isolate RTL-to-emulation mismatches across design, interface, timing, and test-environment layers.

Dynamic Power Profiling Framework for SoC RTL Validation

- Developed a Python framework to parse VCD switching activity, compute switching factors, and estimate dynamic power across RTL execution and datapath blocks under multiple workload scenarios.
- Generated IP-level reports for clock gating and bus utilization, with modeled results showing 88% correlation to available post-silicon measurements.
- Integrated regression hooks to identify power regressions and compare performance-per-watt across builds and benchmark configurations.

DFT Scan and BIST Verification

- Developed and verified scan-test and built-in self-test logic for digital designs, generating ATPG patterns and analyzing fault-simulation results with Synopsys TetraMAX.
- Achieved 95% fault detection in the documented test flow and used simulation/debug outputs to identify coverage gaps and validate test behavior.

FPGA-Based Neural Network Video Processor

- Designed a Verilog real-time inference pipeline with parallel MAC units, bias-addition logic, control FSMs, and ROM-based activation functions for RGB pixel classification.
- Achieved 30 FPS and greater than 95% classification accuracy through RTL simulation, synthesis, timing analysis, and on-board debug using Vivado, ModelSim, and Xilinx ILA.
- Evaluated throughput, resource utilization, and power tradeoffs while automating build and test-generation flows with Tcl and Make.

Power-Aware Inference Optimization for Edge AI on PIM SoCs

- Modeled accuracy, latency, energy, quantization, dimensionality, and analog non-ideality tradeoffs for HDC inference on analog compute-in-memory accelerators using Python and MATLAB.
- Demonstrated up to 42% modeled energy savings with less than 3% accuracy loss and documented architecture constraints for system integration and verification.

VERIFICATION ALIGNMENT

- Verification lifecycle: verification planning, RTL simulation, reusable testbenches, BFM/transactor-based testing, FPGA emulation, regression execution, waveform analysis, failure triage, and CDC/timing debug.
- Verification tooling: SystemVerilog, Python, embedded C, Tcl, Bash, Make, VCS, ModelSim, Xcelium, TetraMAX, Vivado, and Xilinx ILA.
- Hardware-software interaction: firmware-facing embedded C models, ARM embedded Linux, AXI, PCIe, I2C, LVDS, interrupts, and simulation-to-emulation correlation.
- Cross-functional execution: technical design reviews and collaboration across RTL, architecture, verification, FPGA, software, data-science, and embedded-systems activities.

TARGET ROLE

ASIC Verification Engineer – NVIDIA | SoC/GPU Verification, Memory Subsystems, SystemVerilog, and Hardware-Firmware Interaction